Deterministic and Stochastic Replicator Dynamics: Stability, Cycles, and Strategic Evolution

Applied Mathematics / Evolutionary Game Theory — Project Report

Rajveer Vora
Roll No. BSD-CC-2416
rajveeruniversity@gmail.com
Indian Statistical Institute, Kolkata

April 2026

Abstract

We combine analytical stability theory, stochastic perturbation analysis, and a financial market application to study how replicator dynamics captures equilibrium selection, coexistence, and regime transition under strategic adaptation. We analyse four payoff structures: Prisoner's Dilemma, Hawk–Dove, Rock–Paper–Scissors (RPS), and a stylized market microstructure game. Fixed points are classified via Jacobian linearisation and eigenvalue analysis. In RPS, the conserved quantity $H(\mathbf{x}) = x_1 x_2 x_3$ is verified numerically, while stochastic perturbations show that even weak noise destroys invariant level sets and induces boundary exploration absent in the deterministic system. In the market application, replicator dynamics illustrates how mean-reversion can dominate under crowding costs, and sensitivity analysis of the momentum self-payoff reveals a sharp equilibrium transition between momentum- and mean-reversion-dominated regimes.

1. Introduction

Evolutionary game theory (EGT) replaces perfectly rational agents with a population-level selection process: strategies whose payoff exceeds the population average proliferate, while below-average strategies decline. Originating in Maynard Smith and Price [1], the framework now spans economics, sociology, and quantitative finance [6].

The central model is the *replicator equation*, formalising the intuition that *strategies outperforming the average grow*. Two extensions enrich the deterministic baseline. *Stochastic* replicator dynamics model finite-population fluctuations, where noise can destroy conserved quantities and push trajectories to extinction boundaries unreachable by the ODE. In *financial markets*, trading strategies compete for returns as evolutionary archetypes, with market composition evolving by relative fitness.

This report: (i) derives and implements the replicator ODE for four canonical payoff structures; (ii) classifies all fixed points via Jacobian eigenvalue analysis; (iii) quantifies noise-induced effects via stochastic simulation; and (iv) applies the framework to a market microstructure game with parameter sensitivity analysis.

A key geometric feature of replicator dynamics is that the population state evolves on the probability simplex

$$\Delta^{n-1} = \left\{ \mathbf{x} \in \mathbb{R}^n : \; x_i \geq 0, \; \sum_{i=1}^{n} x_i = 1 \right\}.$$

This constraint reflects conservation of total population mass: strategies may gain or lose share, but the total population remains normalized. Geometrically, the simplex provides a natural phase

space in which boundary points correspond to strategy extinction and interior points represent coexistence. This geometry plays a central role in the qualitative behaviour of replicator systems, particularly in distinguishing stable equilibria, cyclic orbits, and stochastic boundary excursions.

2. Mathematical Framework

2.1. State space and notation

Consider a population choosing among n pure strategies. The state is a vector of strategy frequencies

$$\mathbf{x} = (x_1, \ldots, x_n) \in \Delta^{n-1} := \Big\{\mathbf{x} \in \mathbb{R}^n : x_i \geq 0,\ \textstyle\sum_i x_i = 1\Big\}. \tag{1}$$

The payoff structure is encoded in $A \in \mathbb{R}^{n\times n}$, where A_{ij} is the expected payoff to strategy i when matched against strategy j. The *fitness* of strategy i and the *average population fitness* are

$$f_i(\mathbf{x}) = (A\mathbf{x})_i, \qquad \bar{f}(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}. \tag{2}$$

2.2. Replicator equation

$$\dot{x}_i = x_i\big[f_i(\mathbf{x}) - \bar{f}(\mathbf{x})\big] = x_i\big[(A\mathbf{x})_i - \mathbf{x}^\top A\mathbf{x}\big], \qquad i = 1, \ldots, n. \tag{3}$$

Proposition 1 (Simplex invariance)**.** *If* $\mathbf{x}(0) \in \Delta^{n-1}$*, then* $\mathbf{x}(t) \in \Delta^{n-1}$ *for all* $t \geq 0$*.*

Proof. $\frac{d}{dt}\sum_i x_i = \sum_i x_i(f_i - \bar{f}) = \bar{f} - \bar{f} = 0$. Non-negativity is preserved because $\dot{x}_i|_{x_i=0} = 0$. □

Remark 1. *Every Nash equilibrium is a fixed point of* (3)*. An* evolutionarily stable strategy *(ESS) corresponds to an asymptotically stable fixed point.*

Interpretation. The replicator equation can be derived from several independent micro-foundations: imitation dynamics (agents copy better-performing neighbours), best-response adjustment in continuous time, and the limit of finite-population Moran processes as population size grows [4]. Despite its mechanistic simplicity, the equation exhibits the full spectrum of long-run behaviours—convergence to corners, interior fixed points, and persistent cycles—depending entirely on the payoff matrix A. The subsequent sections illustrate each regime through canonical examples, with each game chosen to isolate a distinct qualitative mechanism.

3. Canonical Games

3.1. Prisoner's Dilemma

The Prisoner's Dilemma (PD) involves Cooperate (C) and Defect (D) with

$$A_{\mathrm{PD}} = \begin{pmatrix} R & S \\ T & P \end{pmatrix}, \qquad T > R > P > S, \tag{4}$$

using $T = 5$, $R = 3$, $P = 1$, $S = 0$. Since $T > R$ and $P > S$, Defection strictly dominates. All trajectories converge monotonically to $\mathbf{x}^* = (0, 1)$ (Figure 1), with speed proportional to the fitness gap $T - R$.

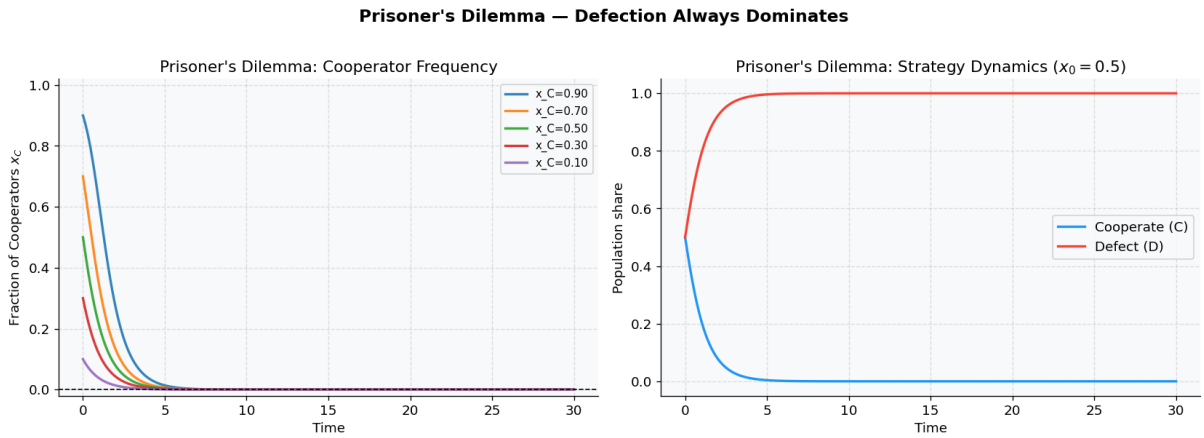


Figure 1. Prisoner's Dilemma. *Left*: cooperator frequency from five initial conditions, all converging to zero. *Right*: both strategies from $x_C(0) = 0.5$. Defection is the unique ESS.

3.2. Hawk–Dove

In the Hawk–Dove game, agents compete for a resource of value V at fighting cost $C > V$:

$$A_{\text{HD}} = \begin{pmatrix} (V-C)/2 & V \\ 0 & V/2 \end{pmatrix}. \tag{5}$$

Setting $\dot{x}_H = 0$ yields the unique interior ESS

$$x_H^* = \frac{V}{C} = \frac{4}{6} = \tfrac{2}{3}. \tag{6}$$

Figure 2 confirms global convergence from all interior starting points; the semi-log deviation plot shows exponential approach with residual error below 10^{-6}.

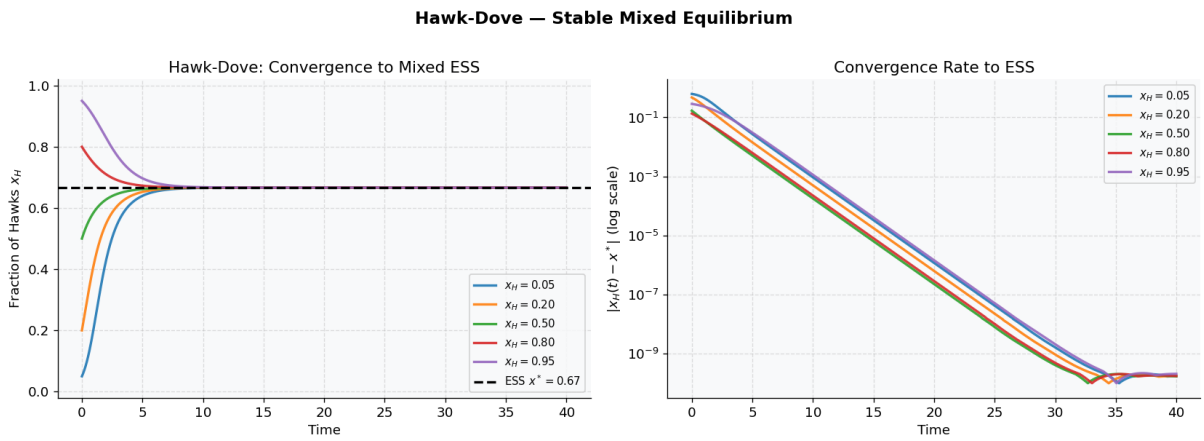


Figure 2. Hawk–Dove dynamics. *Left*: hawk frequency converging to $x^* = 2/3$ (dashed). *Right*: log-scale deviation confirming exponential convergence.

3.3. Rock–Paper–Scissors

The zero-sum RPS payoff matrix is

$$A_{\text{RPS}} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \tag{7}$$

with unique interior Nash equilibrium $\mathbf{x}^* = (1/3, 1/3, 1/3)$.

Theorem 1 (RPS Conserved Quantity)**.** *For the Rock–Paper–Scissors payoff matrix A_{RPS}, the quantity*

$$H(\mathbf{x}) = x_1 x_2 x_3$$

is conserved along every interior trajectory of the replicator dynamics.

Proof. Taking logarithmic derivatives,

$$\frac{\dot{H}}{H} = \sum_{i=1}^{3} \frac{\dot{x}_i}{x_i}.$$

Using the replicator equation,

$$\frac{\dot{x}_i}{x_i} = f_i - \bar{f},$$

so

$$\frac{\dot{H}}{H} = \sum_{i=1}^{3} (f_i - \bar{f}).$$

For the antisymmetric payoff matrix A_{RPS}, we have $\sum_i (Ax)_i = 0$ and $\bar{f} = \mathbf{x}^\top A\mathbf{x} = 0$. Hence,

$$\frac{\dot{H}}{H} = 0,$$

which implies $\dot{H} = 0$. □

Conservation of H confines trajectories to closed level sets; the orbit amplitude depends on the initial distance from the interior equilibrium $\mathbf{x}^*$.

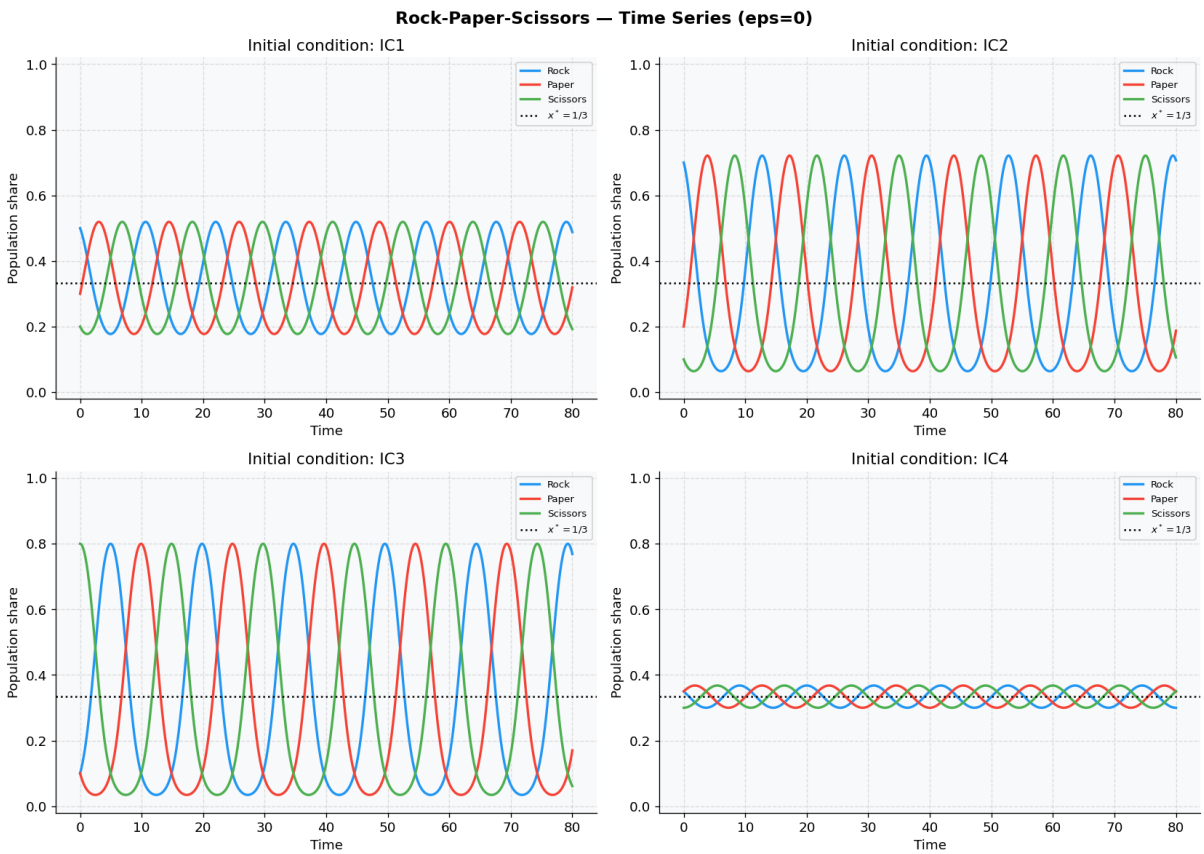
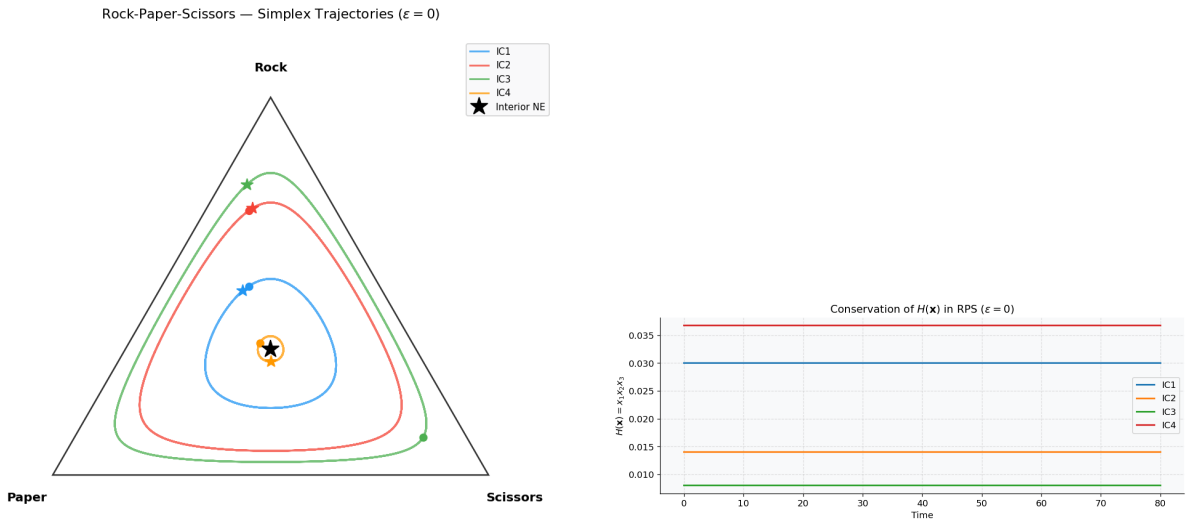


Figure 3. RPS time series for four initial conditions.



(a) Simplex trajectories on invariant level sets. **(b)** Conservation of $H(\mathbf{x}) = x_1x_2x_3$.

Figure 4. Rock–Paper–Scissors dynamics. Left: trajectories remain on closed simplex orbits, with amplitude increasing with initial distance from the interior equilibrium $\mathbf{x}^*$. Right: the invariant $H(\mathbf{x}) = x_1x_2x_3$ remains numerically constant (within 0.01% drift), confirming conservation.

4. Stability Analysis

4.1. Linearisation and Jacobian

Let $\mathbf{x}^*$ be a fixed point of (3). Writing $\mathbf{x} = \mathbf{x}^* + \boldsymbol{\epsilon}$, differentiating the vector field componentwise yields the analytical Jacobian:

$$J_{ij} = \left.\frac{\partial \dot{x}_i}{\partial x_j}\right|_{\mathbf{x}^*} = x_i^*\Big(A_{ij} - (A\mathbf{x}^*)_j - (A^\top\mathbf{x}^*)_j\Big) + \delta_{ij}\big(f_i(\mathbf{x}^*) - \bar{f}(\mathbf{x}^*)\big)\,. \tag{8}$$

Because the state lives on a $(n-1)$-dimensional manifold, J has one structural zero eigenvalue from $\sum_i \epsilon_i = 0$. Stability is determined by the remaining $n-1$ eigenvalues:

- **Stable attractor (ESS):** all non-trivial $\mathrm{Re}(\lambda) < 0$.
- **Unstable:** any non-trivial $\mathrm{Re}(\lambda) > 0$.
- **Neutral center:** all non-trivial λ purely imaginary.

4.2. Results

Table 1 summarises the classification; Figure 5 shows eigenvalue positions in the complex plane.

Table 1. Fixed-point classification. The structural zero eigenvalue is excluded.

Game	Fixed point	Non-trivial λ	Max Re(λ)	Classification
Prisoner's Dilemma	$(0, 1)$	-2.0	-2.0	Stable attractor (ESS)
Hawk–Dove	$(2/3,\ 1/3)$	-0.44	-0.44	Stable attractor (ESS)
Rock–Paper–Scissors	$(1/3, 1/3, 1/3)$	$\pm 1.0\,i$	0.0	Neutral center

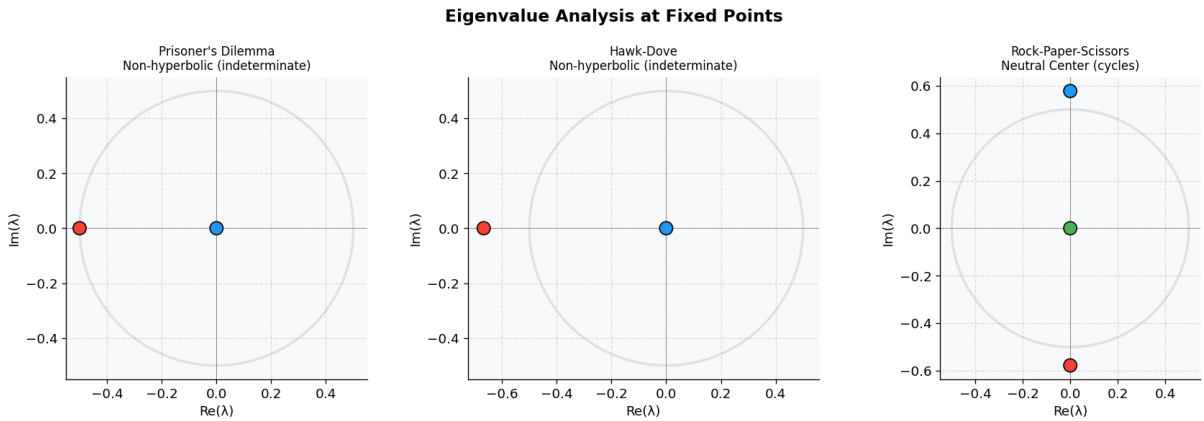


Figure 5. Eigenvalues in the complex plane for the three canonical games. PD and Hawk–Dove eigenvalues lie on the negative real axis (stable attractors). RPS eigenvalues are purely imaginary (neutral center), consistent with perpetual cycling.

The PD equilibrium has a large spectral gap ($|\lambda| = 2.0$), reflecting strict dominance. The Hawk–Dove ESS has a smaller gap ($|\lambda| = 0.44$), consistent with slower convergence (Figure 2). The purely imaginary RPS spectrum algebraically confirms neutral stability and the existence of the conserved invariant H.

5. Stochastic Replicator Dynamics

5.1. Formulation

Finite populations introduce stochasticity, modelled by the Itô SDE

$$dx_i = x_i\big(f_i - \bar{f}\big)\,dt + \sigma\, x_i(1 - x_i)\,dW_t, \tag{9}$$

where $\{W_t\}$ is standard Brownian motion and $\sigma > 0$ is the noise intensity. The diffusion $\sigma x_i(1 - x_i)$ vanishes at the boundaries, preserving simplex geometry. Equation (9) is integrated via the Euler–Maruyama scheme:

$$x_i^{(k+1)} = x_i^{(k)} + x_i^{(k)}\big(f_i - \bar{f}\big)\Delta t + \sigma\, x_i^{(k)}(1 - x_i^{(k)})\sqrt{\Delta t}\, Z_i^{(k)}, \quad Z_i^{(k)} \sim \mathcal{N}(0, 1), \tag{10}$$

followed by clipping and simplex renormalisation at each step.

5.2. Effects on RPS dynamics

Table 2 shows excursion statistics over $T = 750$.

Table 2. Standard deviation of strategy frequencies for deterministic and stochastic RPS, $T = 750$.

Noise level	Rock	Paper	Scissors
Deterministic ($\sigma = 0$)	0.121	0.121	0.122
$\sigma = 0.02$	0.270	0.268	0.269
$\sigma = 0.05$	0.303	0.304	0.303
$\sigma = 0.10$	0.332	0.336	0.331

Figures 6 and 7 reveal three qualitative effects. *Amplitude inflation*: stochastic paths explore a far larger region of the simplex. *Boundary hugging*: at $\sigma = 0.10$, trajectories approach the simplex

edges, driving one strategy near extinction. *Invariant breakdown*: stochastic perturbations destroy the invariant level sets of $H(\mathbf{x})$ and induce diffusive boundary exploration, reflecting the fundamental fragility of neutral stability under noise.

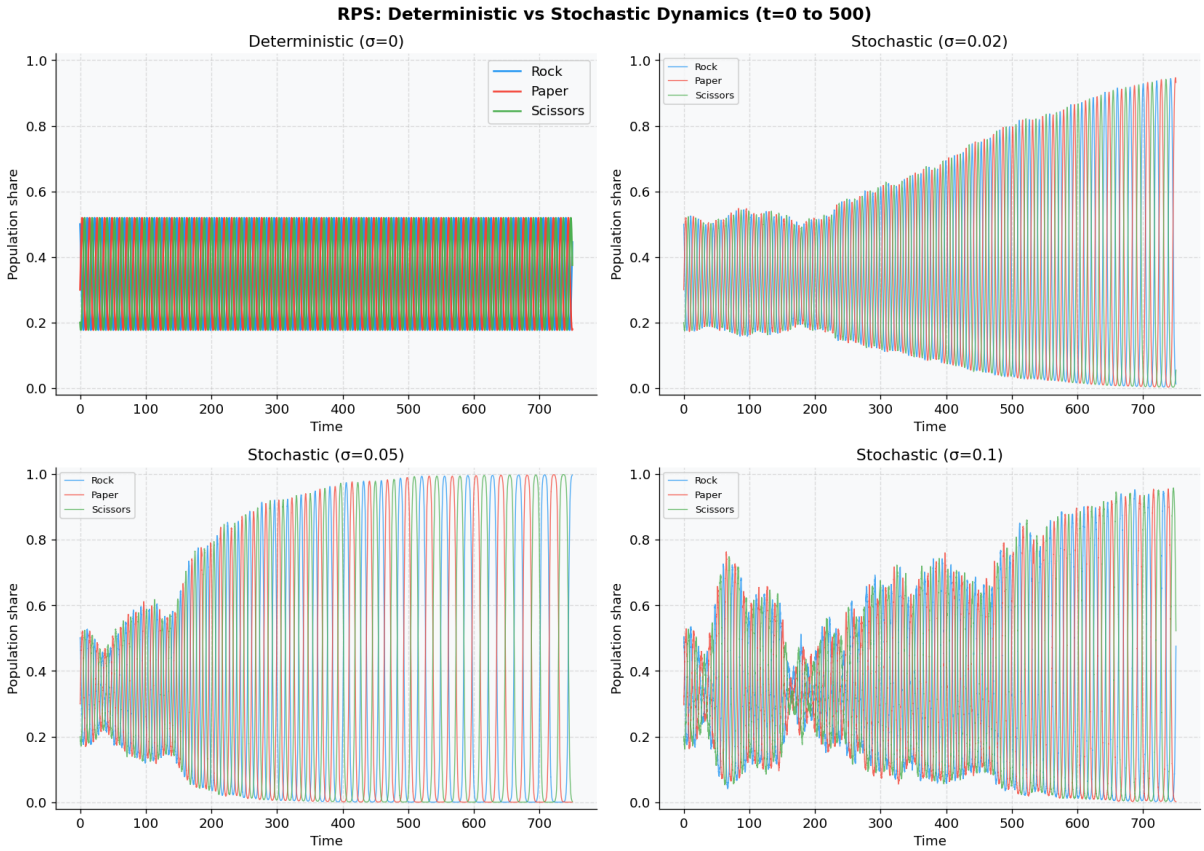


Figure 6. RPS time series ($T = 750$). The deterministic system maintains constant amplitude throughout. At $\sigma = 0.02$ amplitude grows slowly; at $\sigma = 0.05$–0.10 the system reaches boundary-level excursions with near-extinction events visible at $\sigma = 0.10$.

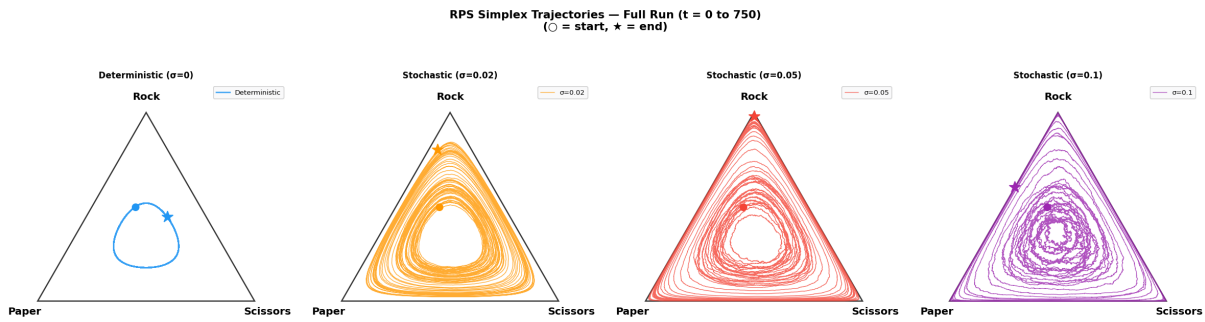


Figure 7. Simplex trajectories from $(0.5, 0.3, 0.2)$ over $T = 750$. Deterministic: clean closed orbit. Stochastic: noise destroys invariant level sets and induces diffusive boundary exploration, with extent increasing monotonically in σ.

6. Market Microstructure Application

6.1. Model

We apply the replicator framework to a stylised market with three trading archetypes: *Momentum* (M), *Mean-Reversion* (MR), and *Noise* (N). Momentum and mean-reversion represent competing alpha-extraction mechanisms that have been empirically documented in equity markets [6], while noise traders model uninformed order flow providing liquidity but earning zero expected return. Their relative dominance depends on the market composition, making this an inherently evolutionary problem. The payoff matrix is

$$A_{\text{mkt}} = \begin{pmatrix} -0.5 & 1.5 & 0.5 \\ 1.0 & 0.2 & 0.3 \\ 0.0 & 0.0 & 0.0 \end{pmatrix}. \tag{11}$$

$A_{00} = -0.5$ captures the crowded-trade effect; $A_{10} = 1.0$ reflects mean-reversionists exploiting momentum overshooting; $A_{11} = 0.2$ gives moderate gains from mean-reversion self-interaction; $A_{2\cdot} = 0$ models zero expected return for noise traders.

6.2. Dynamics and equilibrium

Figure 8 confirms rapid elimination of noise traders from all four representative initial compositions, with the system settling to the Momentum–Mean-Reversion edge near $\mathbf{x}^* \approx (0.46, 0.54, 0)$.

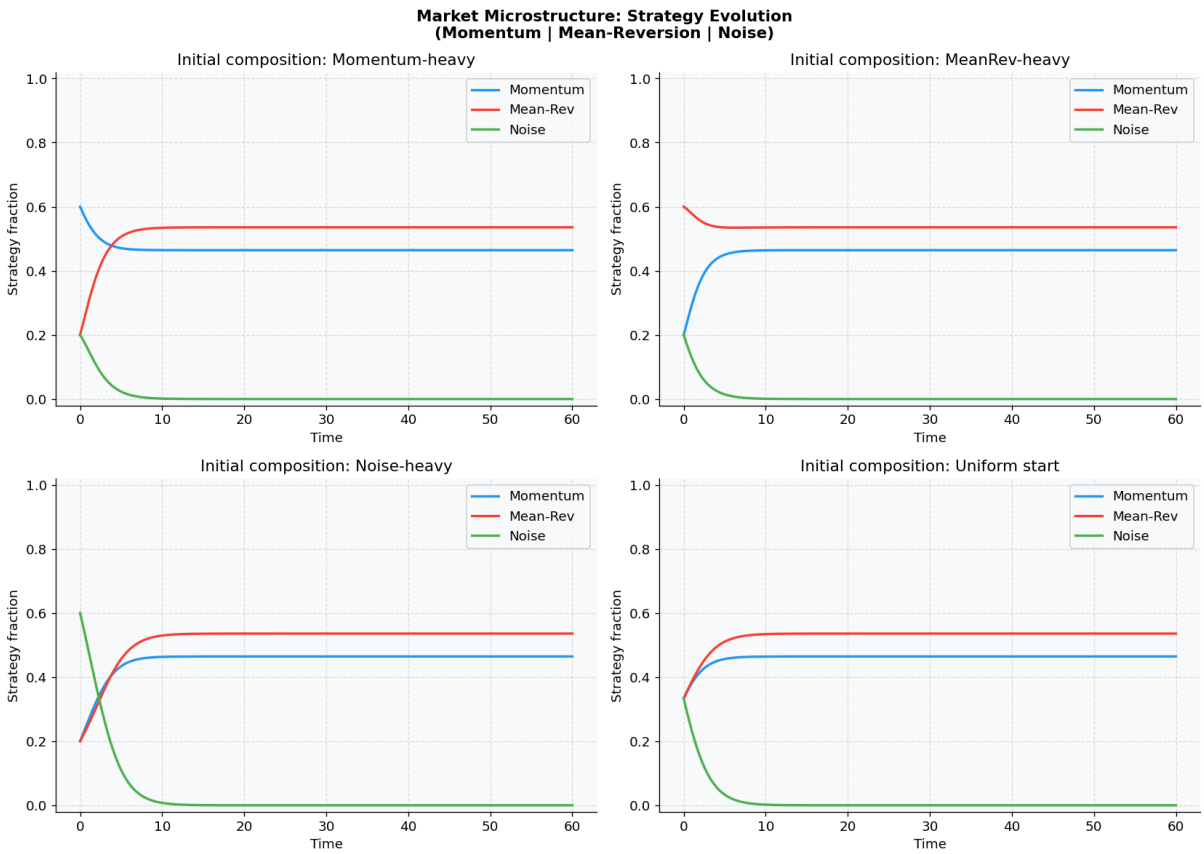


Figure 8. Strategy evolution from four initial compositions. Noise traders are eliminated within $t \approx 10$ in all cases. Momentum and Mean-Reversion converge to a stable split near $(0.46, 0.54, 0)$.

Figure 9 presents the complete phase portrait. The vector field (centre panel) shows flow directed away from the Momentum and Noise vertices toward $\mathbf{x}^*$, confirming the attractor geometry.

All 18 randomly sampled interior initial conditions converge to the same point (right panel), confirming global attraction on the simplex interior.

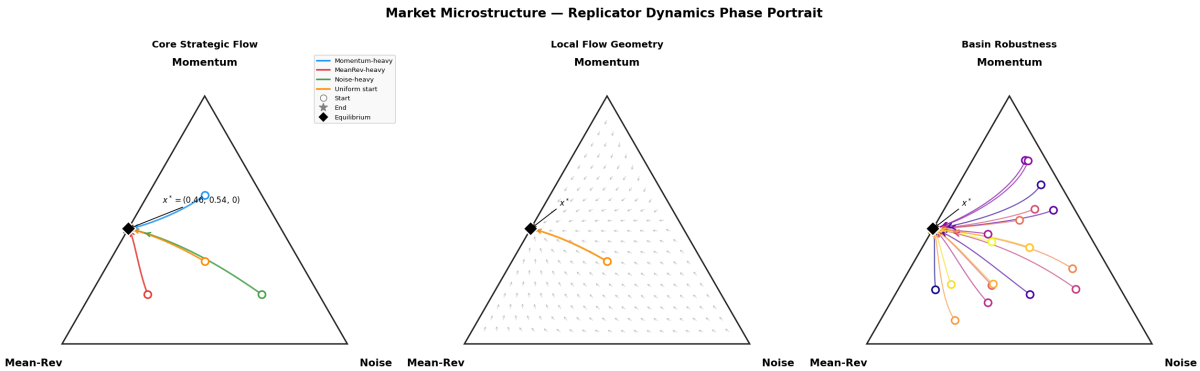


Figure 9. Market microstructure phase portrait. *Left*: representative convergence from four initial compositions. *Centre*: local vector field confirming attractor geometry. *Right*: basin robustness — 18 random starts all converge to $\mathbf{x}^*$.

7. Sensitivity Analysis

The self-payoff A_{00} governs returns when momentum traders interact with one another—a direct proxy for crowding costs. We vary $A_{00} \in [-1.5, 1.0]$, holding all other entries fixed, and record the long-run equilibrium composition.

Figure 10 reveals a clear *regime transition* near $A_{00} \approx 0.3$. Below this threshold, mean-reversion dominates and holds a majority share. Above it, momentum's self-payoff sustains a growing equilibrium fraction, eventually displacing mean-reversion entirely near $A_{00} = 1.0$. Noise traders are extinguished for all parameter values, confirming that zero-return strategies are evolutionarily unstable against any strategy with a positive payoff.

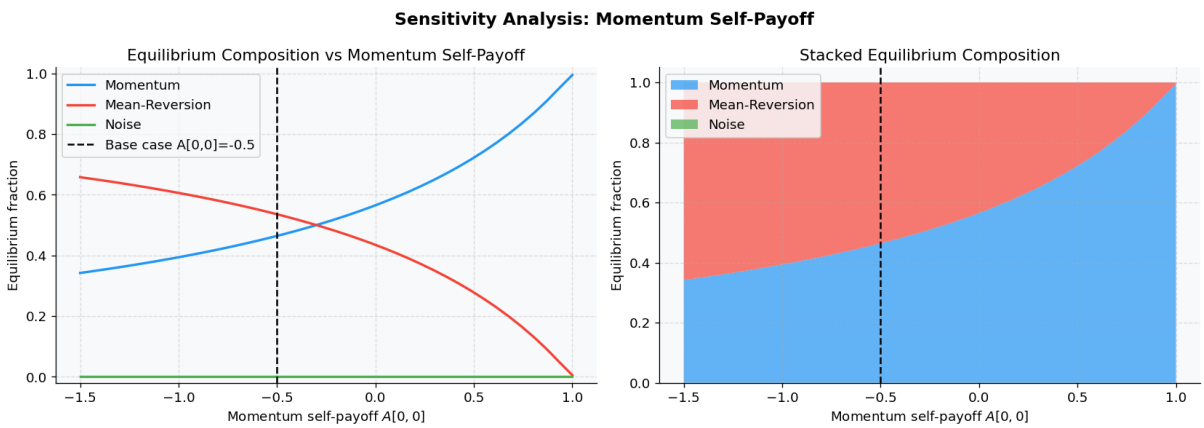


Figure 10. Equilibrium composition as a function of A_{00}. The dashed line marks the base case ($A_{00} = -0.5$), which lies in the mean-reversion-dominated regime consistent with empirical evidence on alpha decay. A regime transition near $A_{00} \approx 0.3$ separates momentum-dominated from mean-reversion-dominated equilibria.

The transition is sharp: a modest increase in A_{00} past the threshold produces a large shift in equilibrium composition. This non-linearity reflects the underlying frequency-dependence of the replicator equation: as momentum's self-interaction becomes less costly, the crowding-out mechanism weakens and a self-reinforcing compositional shift occurs.

8. Conclusion

This report has demonstrated that the replicator equation provides a flexible, analytically tractable framework across qualitatively distinct strategic regimes. The three canonical games span the full behavioural spectrum: strict dominance (PD) drives the system to a corner; frequency-dependent selection (Hawk–Dove) sustains a globally stable interior ESS; cyclic zero-sum competition (RPS) produces neutral orbits governed by a conserved quantity. Eigenvalue analysis provides rigorous classification consistent throughout with the simulation evidence.

Stochastic extensions establish that neutral stability is fragile: weak noise amplifies excursions, breaks the RPS invariant, and causes boundary exploration qualitatively absent from the deterministic system.

The market microstructure application illustrates how mean-reversion can dominate under crowding costs and noise traders are eliminated, while the sensitivity analysis identifies a sharp regime transition in the self-payoff parameter.

Limitations. Several simplifying assumptions limit the scope of these results. The payoff matrix A is fixed and does not respond to changing market conditions or agent learning. All analyses assume an infinite, well-mixed population, abstracting away finite-size effects and network structure. The stochastic model uses independent per-strategy noise rather than a correlated market-wide shock. These assumptions are standard in the EGT literature but would need to be relaxed for empirical application.

Future directions. Natural extensions include *mutation–selection dynamics* to prevent fixation at corners; *adaptive payoffs* where A depends on $\mathbf{x}$; *finite-population models* such as the Moran process; and *spatial networks* supporting cooperation through local interaction.

References

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A. Implementation

All simulations used Python 3.10 with `numpy`, `scipy`, `matplotlib`, and `pandas`. The full notebook is provided as a supplementary file.

A.1 Payoff matrices

```python
def prisoners_dilemma(R=3, P=1, T=5, S=0):
    return np.array([[R, S], [T, P]], dtype=float)

def hawk_dove(V=4, C=6):
    return np.array([[(V-C)/2, V], [0, V/2]], dtype=float)

def rock_paper_scissors(eps=0.0):
    return np.array([[ eps, -1,  1],
                     [  1, eps, -1],
                     [ -1,  1, eps]], dtype=float)

def market_microstructure():
    return np.array([[-0.5, 1.5, 0.5],
                     [ 1.0, 0.2, 0.3],
                     [ 0.0, 0.0, 0.0]], dtype=float)
```

A.2 Deterministic solver

```python
def replicator_rhs(x, t, A):
    x = np.clip(x, 0, 1); x /= x.sum()
    f = A @ x
    return x * (f - x @ f)

def simulate(A, x0, t_end=50, n_steps=5000):
    x0 = np.clip(x0, 1e-8, None); x0 /= x0.sum()
    t = np.linspace(0, t_end, n_steps)
    sol = odeint(replicator_rhs, x0, t, args=(A,), rtol=1e-8, atol=1e-10)
    sol = np.clip(sol, 0, 1)
    return t, sol / sol.sum(axis=1, keepdims=True)
```

A.3 Stochastic solver (Euler–Maruyama)

```python
def stochastic_replicator(A, x0, t_end=50, n_steps=5000,
                          sigma=0.05, seed=None):
    rng = np.random.default_rng(seed)
    x = np.clip(x0, 1e-6, None); x /= x.sum()
    dt = t_end / n_steps; sqrt_dt = np.sqrt(dt)
    sol = np.zeros((n_steps, len(x))); sol[0] = x
    for k in range(1, n_steps):
        f = A @ x; fbar = x @ f
        x = x + x*(f - fbar)*dt \
              + sigma*x*(1-x)*sqrt_dt*rng.standard_normal(len(x))
        x = np.clip(x, 1e-10, None); x /= x.sum()
        sol[k] = x
    return np.linspace(0, t_end, n_steps), sol
```

A.4 Jacobian and stability classification

```python
def classify_fixed_point(A, x_star, tol=1e-8):
    x = np.asarray(x_star, dtype=float)
    eps = 1e-6; n = len(x); J = np.zeros((n, n))
    for j in range(n):
        xf = x.copy(); xb = x.copy()
        xf[j] += eps; xb[j] -= eps
        for v in (xf, xb):
            v[:] = np.clip(v, 0, None); v /= v.sum()
        ff = xf*(A@xf - xf@A@xf); fb = xb*(A@xb - xb@A@xb)
```

```
        J[:, j] = (ff - fb) / (2*eps)
    eigvals = np.linalg.eigvals(J)
    max_re = eigvals.real.max()
    if   max_re < -tol: label = "Stable attractor"
    elif max_re >  tol: label = "Unstable"
    else:               label = "Neutral center"
    return eigvals, label
```

A.5 Ternary simplex utilities

```
def to_cartesian(x1, x2, x3):
    s = x1 + x2 + x3
    cx = 0.5*(2*x3 + x1) / s
    cy = (np.sqrt(3)/2) * x1 / s
    return cx, cy

def draw_simplex_clean(ax, labels, label_offset=0.10):
    h = np.sqrt(3)/2
    tri = plt.Polygon([[0.5,h],[0,0],[1,0]], fill=False,
                      edgecolor='#222', lw=1.8, zorder=1)
    ax.add_patch(tri)
    ax.text(0.50,  h+label_offset, labels[0],
            ha='center', fontsize=13, fontweight='semibold')
    ax.text(-label_offset, -0.05, labels[1],
            ha='center', fontsize=13, fontweight='semibold')
    ax.text(1+label_offset, -0.05, labels[2],
            ha='center', fontsize=13, fontweight='semibold')
    ax.set_xlim(-0.18, 1.18); ax.set_ylim(-0.12, 1.02)
    ax.set_aspect('equal'); ax.axis('off')
```